

Predicting Individual Productivity from Digital Behavior: A Comparative Analysis of Missing-Value Strategies and Machine Learning Models in Regression

DATA1030 Project Report

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[DATA1030]

GitHub Repository:

<https://github.com/muxinfu-dev/data1030project.git>

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1 Introduction

Understanding how everyday digital habits relate to real productivity has become increasingly important as technology shapes work, study, and personal routines. Prior research has shown that frequent digital interruptions and fragmented attention can significantly degrade task performance and increase cognitive load, highlighting the productivity costs of modern digital environments [1]. This project examines whether an individual’s actual productivity score can be predicted from a range of behavioral and lifestyle features, including social-media engagement, notification exposure, sleep and break patterns, and subjective self-assessments. This formulates the task as a supervised regression problem. The dataset, a real-world-style records from a public Kaggle source [2], reflects plausible behavioral relationships while introducing practical modeling challenges such as substantial missingness in features. Because no prior machine-learning study has been conducted on this specific dataset or problem setup, this project establishes a principled baseline by systematically comparing multiple missing-value strategies and evaluating several regression models using cross-validation, with the goal of identifying a robust and interpretable pipeline capable of revealing which behavioral signals are most strongly associated with productivity outcomes.

2 Exploratory Data Analysis (EDA)

To understand the structure of the data and identify relationships relevant to modeling productivity, I conducted a series of exploratory analyses focusing on the distribution of the target variable, cross-feature interactions, and nonlinear behavioral patterns across demographic groups.

2.1 Distribution of the Target Variable

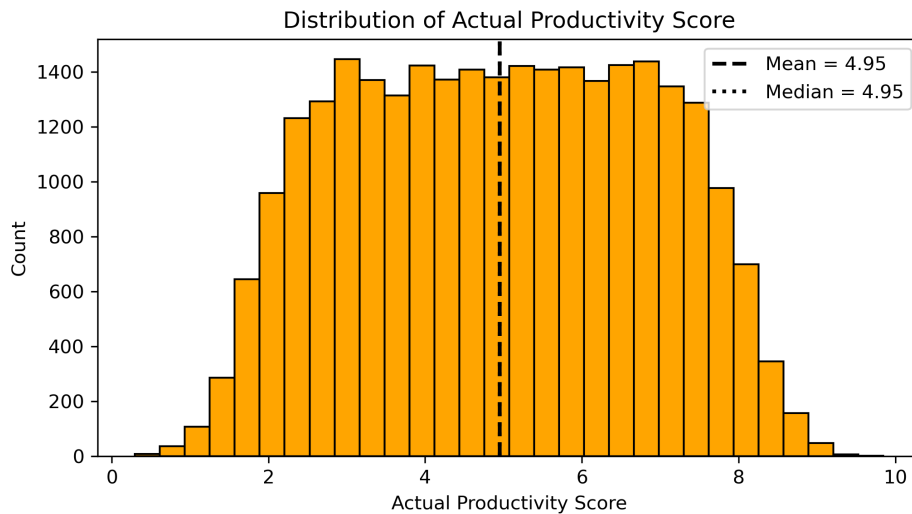


Figure 1: Distribution of `actual_productivity_score`.

Figure 1 shows that most respondents fall within the mid-productivity range. The lack of skew or heavy tails reduces concerns about instability in model fitting and supports the use of linear and nonlinear models without additional normalization of the target.

2.2 Interaction Between Job Type and Social Media Use

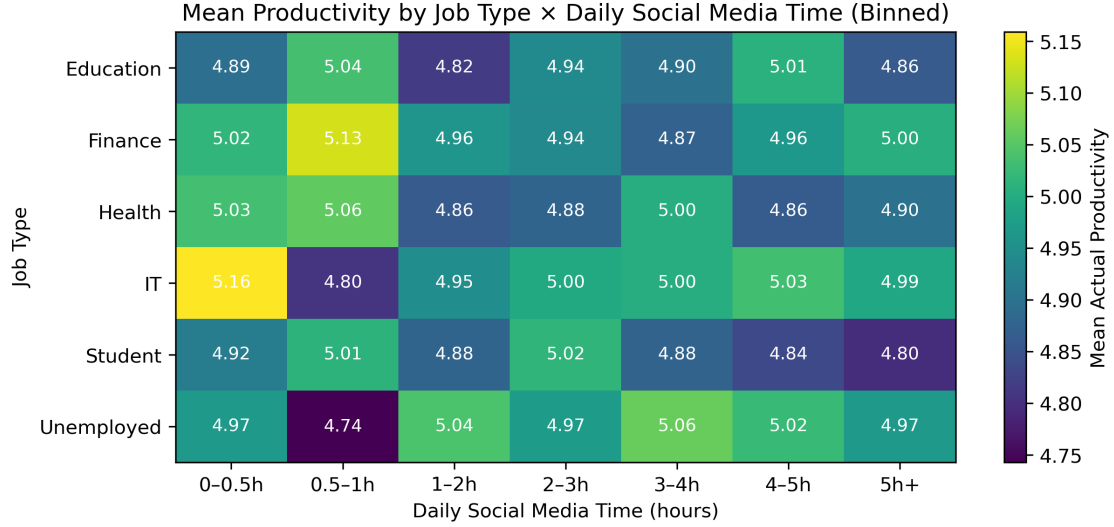


Figure 2: Mean productivity across job types and binned social media usage.

Figure 2 reveals that the relationship between social media use and productivity is highly context-dependent. Productivity for IT and Finance workers varies more noticeably across usage levels, whereas Education and Health remain relatively stable. These differences suggest that the effect of distraction from social media may depend on job structure, responsiveness expectations, or multitasking demands. This motivates the inclusion of interaction terms and tree-based models that can capture heterogeneous effects.

2.3 Nonlinear Behavioral Patterns: Coffee Consumption by Age Group

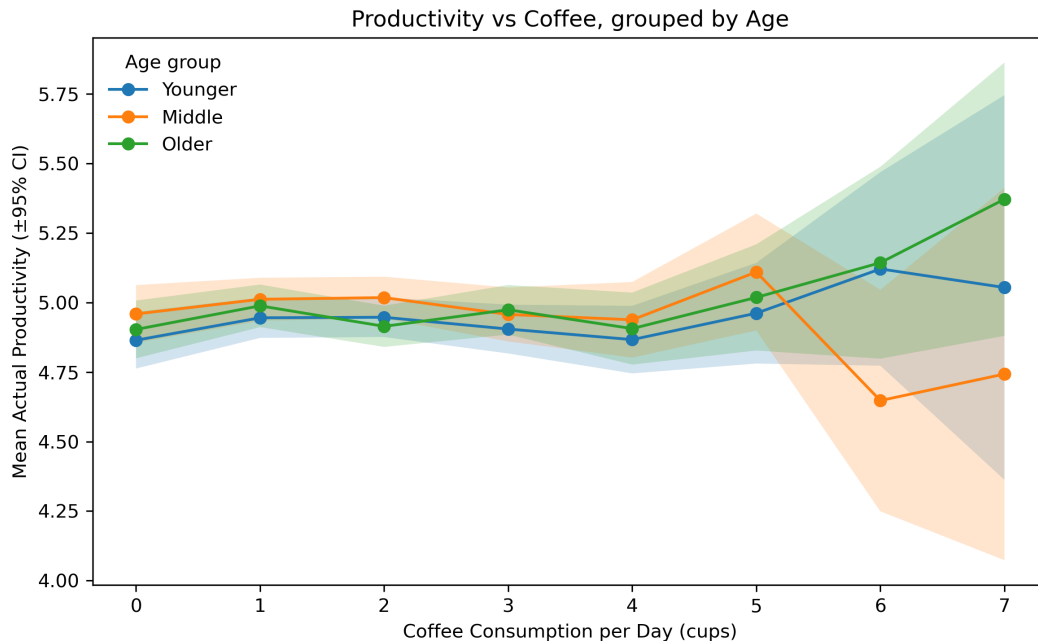


Figure 3: Productivity vs. daily coffee consumption, grouped by age.

In Figure 3, coffee consumption exhibits nonlinear and group-specific patterns. Older participants show a steady increase in productivity with higher coffee intake, whereas middle-aged individuals experience a noticeable decline at high consumption levels. Younger participants remain relatively stable with mild increases. These divergent trends suggest underlying physiological or behavioral differences across age groups and further justify using flexible models capable of capturing nonlinearities and subgroup effects.

3 Methods

3.1 Splitting Strategy and Preprocessing (Part 1)

The dataset consists of independent survey responses, so a random i.i.d. split is appropriate because each observation represents a single, self-contained survey response capturing an individual’s reported digital habits and productivity at one point in time [2]. I therefore split the data into 80% training and 20% test. Before feature engineering or imputation, I inspected missingness patterns to understand whether categorical or numeric handling was required. As shown below, all missingness occurs in numeric rating or continuous features while categorical variables are fully observed.

```
==== Missingness Summary (before preprocessing) ====
Fraction of rows with ANY missing feature: 0.382
```

Number of features with any missing: 6/18 (33.33%)

Top features by missingness:

daily_social_media_time	0.0919
job_satisfaction_score	0.0904
sleep_hours	0.0875
screen_time_before_sleep	0.0739
stress_level	0.0635
perceived_productivity_score	0.0537

Since no categorical missingness is present, numeric features were the only focus of missing-value handling. Feature engineering was performed using training data only: age was bucketed into quantiles, and EDA-motivated interaction terms such as `coffee_x_age` and social-media indicators grouped by different job types (e.g. `sm_time_IT`, `sm_time_Finance`) were added.

3.2 Preprocessing Part 2: Comparing Missing-Value Strategies

Because 38% of rows contain at least one missing numeric value, I compared three strategies using the same XGBoost model and tuning grid to isolate the effect of missing-value handling:

1. **Native missing handling (XGB)**: feed NaNs directly to XGBoost.
2. **IterativeImputer (Random Forest) + XGB**: impute numerics, then scale and encode.
3. **Reduced-feature XGB**: drop frequently missing columns in affected rows.

Strategy	Val RMSE	Val R^2
XGB native missing	0.5020	0.9289
IterativeImputer + XGB	0.5065	0.9276
Reduced-feature XGB	0.5337	0.9196

Table 1: Validation-only comparison of missing-value strategies (model held constant as XGBoost).

Native handling performs best, likely because XGBoost optimally routes missing values during tree construction instead of relying on imputed estimates. Thus, the native-missing design matrix was used for XGBoost, while a full imputation pipeline (IterativeImputer + StandardScaler + OneHotEncoder) was used for models that cannot train with NaNs.

3.3 Model Training and Hyperparameter Tuning

Five models were trained: OLS, Ridge, Random Forest, RBF-kernel SVR, and XGBoost. RMSE was chosen as the primary evaluation metric because it penalizes large prediction

errors, and it has same unit as the target variable, which increases its interpretability. R^2 was also reported for interpretability. GridSearchCV with 5-fold CV was applied to tune hyperparameters. XGBoost used `n_estimators` = 10000 with early stopping to select the optimal number of boosting rounds.

Model	Hyperparameters Searched
XGBoost (native missing)	<code>max_depth</code> $\in \{3, 5, 7\}$; <code>reg_alpha</code> $\in \{0.01, 0.05, 0.1\}$; <code>reg_lambda</code> $\in \{0.1, 1, 10, 30\}$; <code>n_estimators</code> fixed at 10,000 with early stopping
Ridge Regression	$\alpha \in \{0.01, 0.1, 1, 10, 30\}$
SVR (RBF kernel)	$C \in \{1, 5, 10\}$; $\epsilon \in \{0.05, 0.1, 0.3, 0.5\}$
Random Forest	<code>n_estimators</code> $\in \{100, 200, 300, 400\}$; <code>max_depth</code> $\in \{8, 10, 15, 20\}$

Table 2: Hyperparameter search space used in GridSearchCV.

3.4 Quantifying Uncertainty

To measure uncertainty from data splitting and algorithmic randomness, I performed repeated 5-fold cross-validation for all tuned models. Tables 3 and 4 summarize mean performance and variability. XGBoost achieves the lowest validation RMSE, high stability, and a small train validation gap, which supports excellent generalization.

Model	RMSE Train Mean	Train SD	RMSE Val Mean	Val SD	Gap
XGB native missing	0.4862	0.0045	0.5006	0.0098	0.0144
Ridge	0.5049	0.0027	0.5057	0.0106	0.0008
Linear Regression	0.5049	0.0027	0.5057	0.0106	0.0008
Random Forest	0.4634	0.0023	0.5112	0.0107	0.0477
SVR (RBF)	0.4528	0.0027	0.5362	0.0103	0.0835

Table 3: Cross-validated RMSE with uncertainty (train vs. validation).

Model	R^2 Train Mean	Train SD	R^2 Val Mean	Val SD	Gap
XGB native missing	0.9331	0.0012	0.9290	0.0026	0.0041
Ridge	0.9278	0.0007	0.9276	0.0029	0.0003
Linear Regression	0.9278	0.0007	0.9276	0.0029	0.0003
Random Forest	0.9392	0.0006	0.9260	0.0029	0.0132
SVR (RBF)	0.9420	0.0007	0.9186	0.0030	0.0234

Table 4: Cross-validated R^2 with uncertainty (train vs. validation).

Overall, XGBoost with native missing handling demonstrated the best combination of accuracy, stability, and robust generalization across repeated splits.

4 Results

4.1 Baseline Comparison and Overall Predictive Performance

Based on the cross-validated RMSE reported in Table 5, the XGBoost model with native missing handling performs the best since it achieves a mean validation RMSE of 0.5006. All models substantially outperform the baseline RMSE of 1.8797, but XGBoost improves upon it by more than 1.38 points that are equivalent to over 200 standard deviations relative to the baseline’s variability. This magnitude confirms that the improvement cannot be attributed to sampling noise. The validation R^2 results in Table 6 echo this pattern: linear models perform competitively, Random Forest and SVR show mild overfitting, and XGBoost achieves the most stable and highest explanatory power.

On the held-out TEST set (Table 7), XGBoost maintains strong generalization with $\text{RMSE} = 0.5321$ and $R^2 = 0.9214$, far surpassing the baseline. Since the target variable ranges from 0 to 10, an RMSE of approximately 0.53 corresponds to an average prediction error of about 5% of the outcome range. This further supports the high practical accuracy. The true-vs-predicted scatter plot in Figure 4 shows points tightly clustering around the 45-degree line, demonstrating that the model captures the full range of productivity values with little systematic bias.

Model	RMSE Val	Val SD	Improvement vs. Baseline
XGB_native_missing	0.5006	0.0098	1.3791
Ridge	0.5057	0.0106	1.3740
LinearRegression	0.5057	0.0106	1.3740
RandomForest	0.5112	0.0107	1.3686
SVR_rbf	0.5362	0.0103	1.3435
Baseline (mean)	1.8797	0.0061	0.0000

Table 5: Comparison of validation RMSE between the baseline and all trained models.

Model	R^2 Val	Val SD	Improvement vs. Baseline
XGB_native_missing	0.9290	0.0026	0.9293
Ridge	0.9276	0.0029	0.9278
LinearRegression	0.9276	0.0029	0.9278
RandomForest	0.9260	0.0029	0.9263
SVR_rbf	0.9186	0.0030	0.9188
Baseline (mean)	-0.0003	0.0003	0.0000

Table 6: Comparison of validation R^2 between the baseline and all trained models.

Model	RMSE (TEST)	R^2 (TEST)
Baseline (mean)	1.8981	-0.0001
Final XGBoost (native missing)	0.5321	0.9214

Table 7: Baseline vs. final XGBoost model on the held-out TEST set.

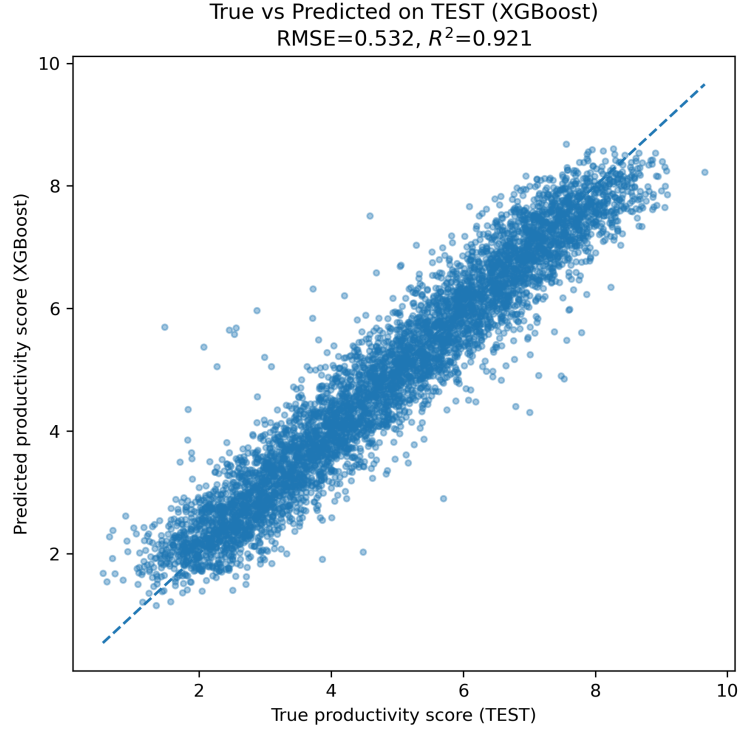


Figure 4: True vs. predicted productivity scores on TEST for the final XGBoost model.

4.2 Global Feature Importance

To understand which variables drive the model’s performance, I examined three XGBoost importance metrics: gain, weight, and cover, as shown in Figures 5–7, along with SHAP global importance in Figure 8. Across all methods, `perceived_productivity_score` and `job_satisfaction_score` dominate by a large margin, indicating that internal psychological assessments are the strongest predictors of actual productivity. Behavioral factors such as `sleep_hours`, `daily_social_media_time`, and `screen_time_before_sleep` consistently rank below them, suggesting that such habits refine rather than determine overall productivity.

Cover-based importance highlights job-specific social-media interaction features such as `sm_time_Finance` and `sm_time_IT`. Although these do not always provide the largest immediate gain, they frequently influence how many observations are routed through different subtrees, thus supporting the EDA finding that social-media effects vary across occupations. Meanwhile, sparse demographic categories and platform preferences contribute little once core psychological and behavioral variables are included.

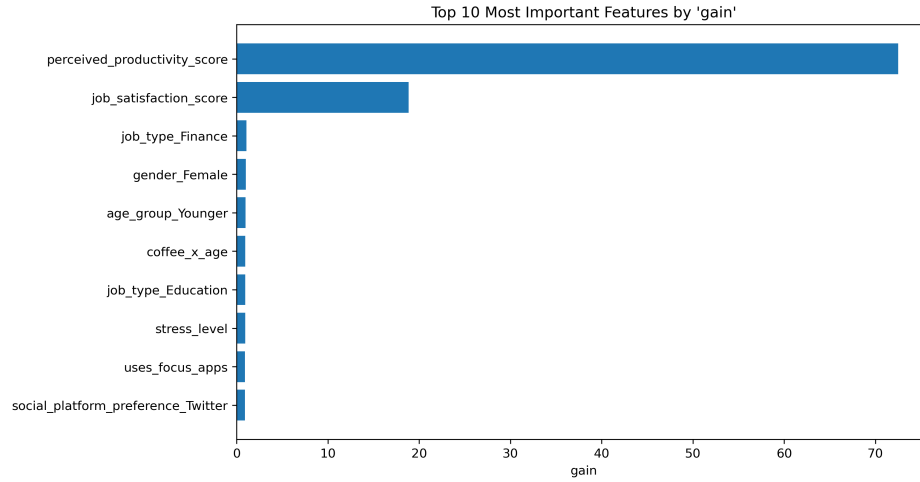


Figure 5: Top 10 global feature importances by XGBoost gain.

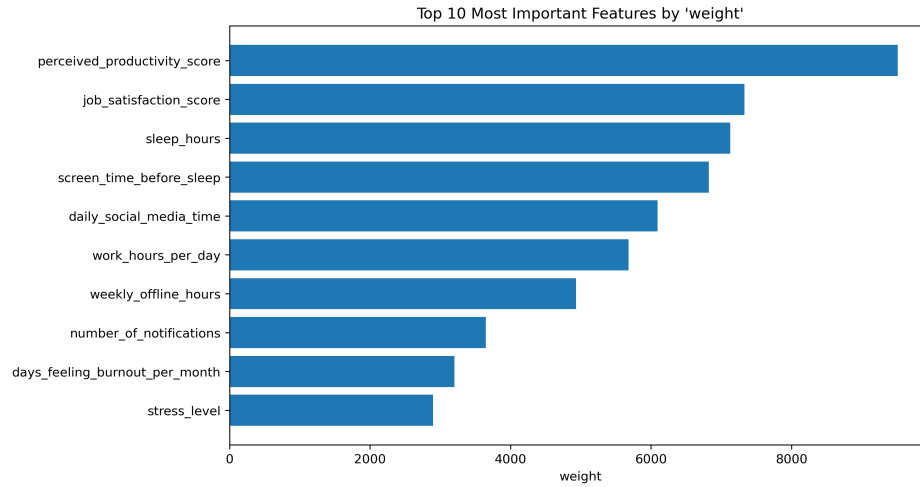


Figure 6: Top 10 global feature importances by XGBoost weight.

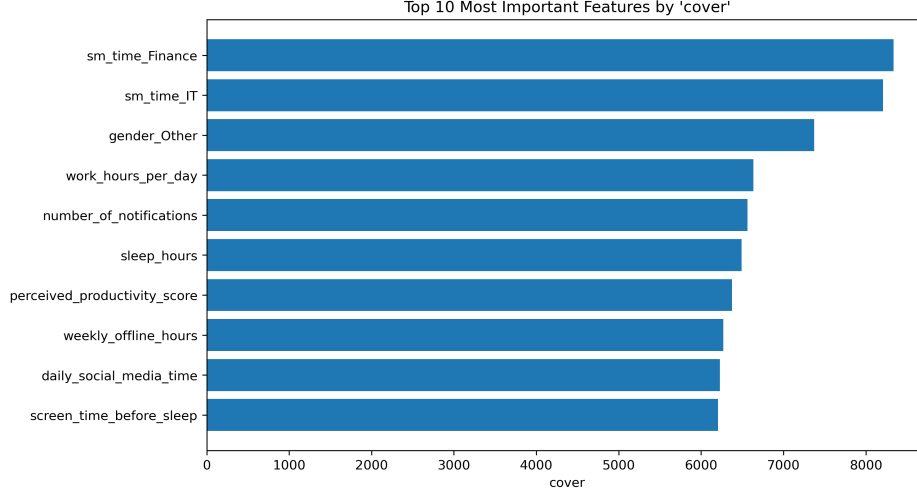


Figure 7: Top 10 global feature importances by XGBoost cover.

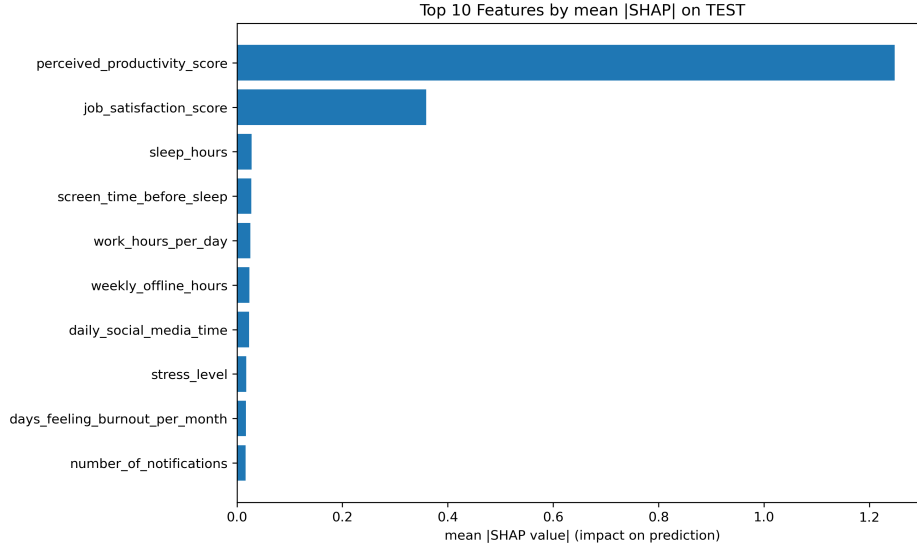


Figure 8: Top 10 features by mean absolute SHAP value on the TEST set.

4.3 Local Feature Importance with SHAP

To examine how global structures translate to individual predictions, Figure 9 shows SHAP force plots for three representative test cases. For a high-productivity user (index 0), strong positive SHAP contributions from `perceived_productivity_score` and `job_satisfaction_score` dominate. For a low-productivity case (index 100), negative psychological scores and the `coffee_x_age` interaction drive the prediction downward. A moderate case (index 200) exhibits compensating effects, with positive perceived productivity offset by late-night screen time.

Overall, the force plots demonstrate that although the same global hierarchy applies across users, each prediction emerges from a distinct combination of behavioral pushes and

pulls.

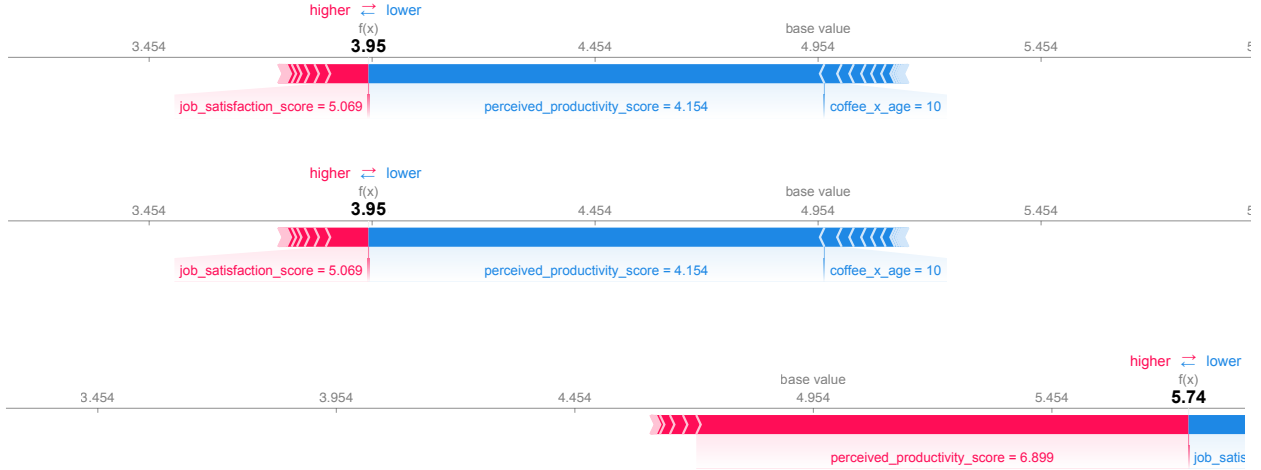


Figure 9: SHAP force plots for three representative TEST individuals (indices 0, 100, and 200).

5 Outlook

Several opportunities exist to strengthen both the predictive performance and interpretability of the current modeling approach. One clear limitation is the heavy reliance on self-reported psychological variables, such as `perceived_productivity_score` and `job_satisfaction_score`, which, while highly predictive, may reflect bias or reverse causality. A natural next step would be to incorporate objective behavioral data (e.g., screen-usage logs, keystroke-based activity measures, or wearable-derived sleep metrics) to provide more stable signals and support stronger causal interpretation.

Modeling improvements could also enhance robustness. Although XGBoost performed best among the tested models, further exploration of nonlinear and individual-specific effects may be beneficial. Techniques such as generalized additive models (GAMs), mixed-effects or hierarchical models, or temporal models (if longitudinal data were collected) could better capture heterogeneity in how habits influence productivity.

Finally, collecting a larger and more diverse dataset would help reduce variance, allow models to generalize across subpopulations, and support more detailed subgroup analyses. Together, these extensions would improve model reliability while offering deeper insight into how digital and lifestyle behaviors shape real productivity.

6 References

References

- [1] Mark, G., Gudith, D., and Klocke, U. The Cost of Interrupted Work: More Speed and Stress. *Communications of the ACM*, 51(4), 2008. Available at: <https://ics.uci.edu/~gmark/chi08-mark.pdf>
- [2] Mashyekhi, M. *Social Media vs Productivity Dataset*. Kaggle, 2023. Available at: <https://www.kaggle.com/datasets/mahdimashayekhi/social-media-vs-productivity>